

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Semester : III

Branch : Electronics and Computer Engg. / Electronics and Computer Science Engg.

Subject Code & Name: BTESC305 Digital Electronics and Microprocessor

Max Marks: 60

Date:11-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Perform following Conversion.	(k1/CO1)	6
a. $(0.1010)_2 = (\dots\dots\dots?)_{10}$		
b. $(249)_{10} = (\dots?)_{16}$		
c. $(3241)_8 = (\dots\dots\dots?)_2$		
B) What are the universal gates? Explain how basic gates can be realized using NAND and NOR gates.	(k3/CO1)	6
C) Elaborate need of coding? Explain error detecting and error correcting codes.	(k2/CO1)	6
Q.2 Solve Any Two of the following.		12
A) Minimize following equation using Kmap	(k1/CO2)	6
a) $F(A,B,C,D) = \sum m(0,1,2,3,4,5,13,15) + d(8,9,10,11)$		
b) $F(A,B,C,D) = \prod M(0,2,4,6,8,10) \cdot d(1,3,5,7)$		
B) Explain the working of half adder and full adder with truth table	(k1/CO2)	6
C) Provide comprehensive explanation of ALU using IC 74181.	(k2/CO2)	6
Q. 3 Solve Any Two of the following.		12
A) Define Flip-Flop? Explain S-R Flip –Flop.	(k2/CO3)	6
B) Draw and explain universal shift register.	(k3/CO3)	6
C) Differentiate between synchronous counter and asynchronous counter.	(k2/CO3)	6
Q.4 Solve Any Two of the following.		12
A) Draw architecture of 8085 microprocessor and explain it in detail.	(k1/CO4)	6
B) Explain the addressing modes of 8085 with suitable examples of each.	(k3/CO4)	6

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| C) Write a program for addition of two 16 bit numbers using 8085. Store lower byte of a result at 1000H memory location and higher byte 1001H. | (k2/CO4) | 6 |
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Q. 5 Solve Any Two of the following.		12
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| A) Compare between 8085 and 8086 microprocessor. | (k1/CO5) | 6 |
| B) Describe 8086 flag register and its functions. | (k3/CO5) | 6 |
| C) Explain hardware and software interrupts of 8086. | (k2/CO5) | 6 |

*** End ***